

NUIQSUT, AK, USA

WMO: 703644

Lat: 70.212N

Lon: 151.002W

Elev: 57

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 27515

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.3	-40.5	-35.4	0.7	-27.2	-34.3	0.8	-27.6	36.7	-3.4	33.6	-1.9	5.8	200	1.014

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	69.8	58.4	64.7	55.6	61.1	54.0	59.1	67.3	56.6	63.7	54.3	60.3	10.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.9	64.2	61.9	52.6	58.9	59.7	50.3	54.2	57.8	25.7	67.4	24.1	63.3	22.7	60.6	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.9	26.6	23.7	-46.2	76.7	5.5	4.0	-50.2	79.6	-53.4	82.0	-56.5	84.2	-60.5	87.2		
(5)			-33.1	63.0	1.2	4.7	-33.9	66.3	-34.6	69.1	-35.3	71.7	-36.1	75.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	-14.5	-12.5	-13.2	3.0	24.0	42.2	49.6	44.3	35.6	20.7	2.0	-9.5
	DBStd	26.12	14.28	15.83	13.50	12.11	9.77	8.51	6.99	7.11	6.51	8.51	12.64	14.12
	HDD50	13130	2000	1750	1961	1409	806	267	97	212	434	908	1440	1846
	HDD65	18453	2465	2170	2426	1859	1271	684	479	643	882	1373	1890	2311
	CDD50	153	0	0	0	0	0	33	83	35	2	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	23	0	0	0	0	0	6	15	2	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	10.5	10.8	11.0	10.8	10.6	10.8	11.3	10.2	9.7	9.9	10.5	10.2	10.1
Precipitation	PrecAvg	8.1	0.4	0.4	0.3	0.2	0.4	0.6	1.2	1.5	1.1	1.0	0.6	0.5
	PrecMax	13.7	1.3	1.1	1.7	0.6	0.8	1.5	2.4	3.7	3.6	2.0	1.5	1.4
	PrecMin	4.6	0.1	0.0	0.0	0.0	0.0	0.1	0.2	0.4	0.2	0.4	0.2	0.1
	PrecStd	1.8	0.3	0.2	0.3	0.2	0.2	0.4	0.5	0.7	0.8	0.4	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	29.2	26.3	36.5	49.6	73.2	75.6	72.0	61.6	40.1	32.0	28.1
		MCWB	29.4	28.0	24.9	34.5	43.3	57.5	60.4	60.7	51.7	38.1	31.2	25.9
	2%	DB	21.2	22.6	19.1	30.0	43.0	66.6	71.6	65.5	54.9	35.4	26.5	21.3
		MCWB	20.3	21.4	18.6	28.9	39.4	55.1	59.2	56.6	48.8	34.1	25.3	19.8
	5%	DB	10.0	17.3	11.6	25.4	38.7	61.6	66.5	61.1	49.5	33.0	22.7	15.4
		MCWB	9.1	16.6	10.9	24.3	36.5	53.0	57.1	54.4	46.1	32.3	21.8	14.7
	10%	DB	3.6	9.9	4.4	20.7	35.8	56.7	62.7	56.4	44.6	30.7	19.5	9.0
		MCWB	3.3	8.9	3.8	19.9	34.3	50.4	55.3	52.2	42.6	30.0	18.7	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	28.2	26.0	34.7	43.7	59.2	63.4	61.3	52.3	38.3	31.1	26.5
		MCDB	30.7	29.2	26.7	36.4	48.4	70.9	74.0	69.5	59.4	40.0	31.8	27.6
	2%	WB	22.0	22.6	19.1	28.9	39.4	55.8	60.1	57.6	49.2	34.4	25.6	20.8
		MCDB	22.8	23.7	19.7	29.7	42.5	65.1	68.9	63.8	55.1	35.1	26.3	22.1
	5%	WB	11.2	17.2	12.0	24.6	36.5	53.4	57.8	54.8	46.1	32.2	21.8	15.0
		MCDB	11.8	18.0	12.7	25.6	38.5	61.0	65.3	59.8	49.6	32.8	22.5	15.8
	10%	WB	4.8	11.1	5.6	19.9	34.3	50.7	55.6	52.3	42.6	30.2	18.8	8.8
		MCDB	5.5	12.1	6.6	20.5	35.5	56.9	61.9	56.4	44.6	30.7	19.6	9.5
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.7	12.9	14.0	9.4	14.2	15.6	12.1	9.3	9.5	13.1	12.6
		MCDBR	21.4	20.2	16.5	14.9	13.2	22.7	23.6	20.9	19.5	9.0	14.3	17.4
		MCWBR	20.7	19.5	16.1	14.1	10.1	13.2	13.0	12.6	13.3	8.2	13.6	16.9
	5% WB	MCDBR	21.9	20.4	16.7	14.7	12.7	22.4	21.8	19.1	19.0	8.6	13.9	17.3
		MCWBR	21.5	19.7	16.4	14.0	9.8	13.6	13.1	12.0	13.0	8.0	13.4	16.8
Clear-Sky Solar Irradiance	taub	N/A	0.202	0.239	0.297	0.326	0.331	0.343	0.335	0.290	0.216	N/A	N/A	
	taud	N/A	2.064	2.236	2.146	2.200	2.404	2.449	2.497	2.536	2.159	N/A	N/A	
	Ebn at Noon	0	172	251	263	269	273	265	253	236	161	0	0	
	Edh at Noon	0	12	21	33	36	30	28	24	17	10	0	0	
All-Sky Solar Radiation	RadAvg	0	115	623	1370	1880	1961	1615	946	477	143	0	0	
	RadStd	0	8	23	61	102	133	126	96	46	19	0	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		+2.05	N/A	N/A	N/A	N/A	N/A	-902	-889	N/A	N/A	

Nomenclature: See separate page